EST II — MATH LEVEL 2 • UNIT 7.01

C Descriptive Statistics, Normal Distribution & Regression

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 A	Q3 A	Q4 A	Q5 A
Q6 A	Q7 A	Q8 B	Q9 C	Q10 B
Q11 C	Q12 B	Q13 A	Q14 A	Q15 D
Q16 C	Q17 B	Q18 D	Q19 B	Q20 D
Q21 D	Q22 B	Q23 A	Q24 B	Q25 C

Descriptive Statistics, Normal Distribution & Regression Practice Exercises

1 MCQ

If $i = \sqrt{-1}$, which of the following is equivalent to the complex expression $\frac{7+4i}{2-i}$?

A $2 + 3i$ ✓

B $2 - 3i$

C $\frac{10+15i}{3}$

D $\frac{18+15i}{5}$

WORKED SOLUTION & VERIFICATION

To simplify the quotient of two complex numbers, multiply both the numerator and the denominator by the complex conjugate of the denominator, which is $2 + i$:

$$\frac{7+4i}{2-i} = \frac{(7+4i)(2+i)}{(2-i)(2+i)}$$

Expand the numerator using the distributive property (FOIL):

$$(7+4i)(2+i) = 14 + 7i + 8i + 4i^2$$

Since $i^2 = -1$:

$$14 + 15i + 4(-1) = 14 - 4 + 15i = 10 + 15i$$

Expand the denominator using the difference of squares:

$$(2-i)(2+i) = 2^2 - i^2 = 4 - (-1) = 5$$

Now, divide each term in the numerator by the denominator:

$$\frac{10+15i}{5} = \frac{10}{5} + \frac{15i}{5} = 2 + 3i$$

Common Misconceptions:
- Choice B occurs from an arithmetic sign error in the imaginary term.
- Choice C occurs if the student mistakenly calculates the denominator as $2^2 - 1^2 = 3$ by forgetting that $-i^2 = -(-1) = +1$.
- Choice D occurs if the student takes $i^2 = 1$, yielding $14 + 4 = 18$ in the numerator.

2 MCQ

The graph of the function $y = f(x)$ is shown below. What is the value of $f(1)$?

- **A** 0 ✓
- **B** 2
- **C** -1
- **D** 3

WORKED SOLUTION & VERIFICATION

The value of $f(1)$ corresponds to the y-coordinate of the point on the graph where the x-coordinate is 1. Looking at the provided graph, the point $(1, 0)$ is marked. Therefore, $f(1) = 0$.

3 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. What are the x-intercepts of the graph of this function?

- **A** **(1, 0) and (3, 0)** ✓
- **B** (0, 1) and (0, 3)
- **C** (-1, 0) and (-3, 0)
- **D** (1, 0) and (-3, 0)

WORKED SOLUTION & VERIFICATION

The x-intercepts of a function occur where the function's value is zero, i.e., $f(x) = 0$. \nWe need to solve the quadratic equation $x^2 - 4x + 3 = 0$. \nThis equation can be factored as $(x - 1)(x - 3) = 0$. \nTherefore, the solutions are $x = 1$ and $x = 3$. \nThis means the x-intercepts are $(1, 0)$ and $(3, 0)$.

4 MCQ

A vector $\mathbf{v}$ is represented by the directed line segment from point $P(1, 2)$ to point $Q(4, 6)$. What are the components of vector $\mathbf{v}$?

- **A** **[3, 4]** ✓
- **B** [5, 8]
- **C** [-3, -4]
- **D** [4, 3]

WORKED SOLUTION & VERIFICATION

The components of a vector $\mathbf{v}$ represented by a directed line segment from point $P(x_1, y_1)$ to point $Q(x_2, y_2)$ are given by $\mathbf{v} = [x_2 - x_1, y_2 - y_1]$. In this case, $P = (1, 2)$ and $Q = (4, 6)$. Therefore, the components of $\mathbf{v}$ are $[4 - 1, 6 - 2] = [3, 4]$.

5 MCQ

A bag contains 12 red marbles and 8 blue marbles. If one marble is drawn at random, what is the probability that the marble is red?

- **A** $\frac{3}{5}$ ✓
- **B** $\frac{2}{5}$
- **C** $\frac{1}{2}$
- **D** $\frac{3}{4}$

WORKED SOLUTION & VERIFICATION

The total number of marbles in the bag is the sum of red and blue marbles: $12 + 8 = 20$ marbles. The number of favorable outcomes (drawing a red marble) is 12. The probability of an event is calculated as the ratio of the number of favorable outcomes to the total number of possible outcomes. Therefore, the probability of drawing a red marble is $P(\text{red}) = \frac{\text{Number of red marbles}}{\text{Total number of marbles}} = \frac{12}{20}$. This fraction can be simplified by dividing both the numerator and the denominator by their greatest common divisor, which is 4: $\frac{12 \div 4}{20 \div 4} = \frac{3}{5}$. Thus, the probability of drawing a red marble is $\frac{3}{5}$.

6 MCQ

Consider an arithmetic sequence where the first term is $a_1 = 5$ and the common difference is $d = 3$. What is the value of the 4th term, a_4?

- **A** 14 ✓
- **B** 17
- **C** 8
- **D** 11

WORKED SOLUTION & VERIFICATION

The formula for the nth term of an arithmetic sequence is given by $a_n = a_1 + (n-1)d$. In this problem, we are given $a_1 = 5$ and $d = 3$. We need to find the 4th term, so $n = 4$. Plugging these values into the formula, we get: $a_4 = 5 + (4-1) \times 3$ $a_4 = 5 + (3) \times 3$ $a_4 = 5 + 9$ $a_4 = 14$. Therefore, the 4th term of the sequence is 14.

7 MCQ

A geometric sequence has $a_1 = 6$ and common ratio $r = 2$. Find a_5.

- **A** 96 ✓
- **B** 192
- **C** 98
- **D** -96

WORKED SOLUTION & VERIFICATION

Applying Sequences & Series principles, $a_5 = a_1 r^{n-1} = 6(2)^4 = 96$. Therefore, the verified correct option is 96.

8 MCQ

The graph of the function $y = f(x) = ax^2 + bx + c$ is shown below. The graph passes through the points $(-1, 5)$, $(1, 1)$, and $(3, 5)$. Determine the value of b.

A 0

B -2 ✓

C 2

D 1

WORKED SOLUTION & VERIFICATION

The general form of the quadratic function is $y = ax^2 + bx + c$. We are given three points that lie on the graph: $(-1, 5)$, $(1, 1)$, and $(3, 5)$.

Substituting these points into the equation, we get a system of three linear equations with three unknowns (a, b, and c):

1. For $(-1, 5)$: $a(-1)^2 + b(-1) + c = 5 \Rightarrow a - b + c = 5$
2. For $(1, 1)$: $a(1)^2 + b(1) + c = 1 \Rightarrow a + b + c = 1$
3. For $(3, 5)$: $a(3)^2 + b(3) + c = 5 \Rightarrow 9a + 3b + c = 5$

We can solve this system of equations. Let's subtract equation (1) from equation (2):

$(a + b + c) - (a - b + c) = 1 - 5$

$2b = -4$

$b = -2$

Alternatively, notice that the y-values for $x = -1$ and $x = 3$ are the same (both are 5). This indicates that the axis of symmetry of the parabola is exactly in the middle of $x = -1$ and $x = 3$. The x-coordinate of the vertex (axis of symmetry) is given by $-\frac{b}{2a}$.

Axis of symmetry: $x = \frac{-1+3}{2} = \frac{2}{2} = 1$

So, $-\frac{b}{2a} = 1$, which implies $b = -2a$.

Now, substitute this into equation (2): $a + b + c = 1$

$a + (-2a) + c = 1$

$-a + c = 1 \Rightarrow c = a + 1$

Substitute $b = -2a$ and $c = a + 1$ into equation (1):

$a - (-2a) + (a + 1) = 5$

$a + 2a + a + 1 = 5$

$4a + 1 = 5$

$4a = 4$

$a = 1$

Now we can find b using $b = -2a$:

$b = -2(1)$

$b = -2$

To confirm, we can also find c: $c = a + 1 = 1 + 1 = 2$. The function is $y = x^2 - 2x + 2$.

Let's check the points:

For $(-1, 5)$: $(-1)^2 - 2(-1) + 2 = 1 + 2 + 2 = 5$ (Correct)

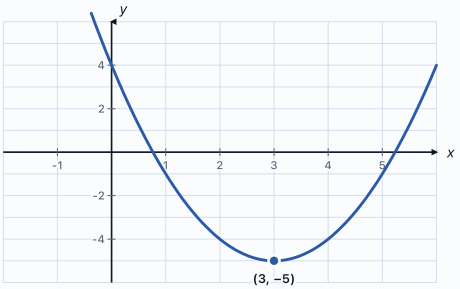
For $(1, 1)$: $(1)^2 - 2(1) + 2 = 1 - 2 + 2 = 1$ (Correct)

For $(3, 5)$: $(3)^2 - 2(3) + 2 = 9 - 6 + 2 = 5$ (Correct)

The value of b is -2.

9 MCQ

The graph of the function $f(x) = x^2 - 6x + k$ is shown below. The graph has a vertex at $(3, -5)$. Determine the value of the parameter k.



A 9

B -4

C 4 ✓

D 1

WORKED SOLUTION & VERIFICATION

The vertex of a parabola in the form $f(x) = ax^2 + bx + c$ occurs at $x = -b/(2a)$. For $f(x) = x^2 - 6x + k$, we have $a = 1$ and $b = -6$. The x-coordinate of the vertex is $-(-6)/(2 \cdot 1) = 6/2 = 3$, which matches the given vertex. The y-coordinate of the vertex is $f(3)$. We are given that the vertex is $(3, -5)$, so $f(3) = -5$. Substituting $x = 3$ into the function: $f(3) = (3)^2 - 6(3) + k = 9 - 18 + k = -9 + k$. Setting this equal to the given y-coordinate of the vertex: $-9 + k = -5$. Solving for k: $k = -5 + 9 = 4$. Thus, the value of the parameter k is 4. This means the function is $f(x) = x^2 - 6x + 4$. The vertex y-coordinate calculation is $f(3) = 3^2 - 6(3) + 4 = 9 - 18 + 4 = -5$. The parameter k is 4, not 1. Common algebraic traps include: 1. Errors in the vertex formula. 2. Incorrectly substituting the vertex x-coordinate back into the function. 3. Sign errors when solving for k. In this case, the correct value of k is 4.

10 MCQ

A translation maps point $A(1, 2)$ to $A'(-4, 2)$ and point $B(3, 5)$ to $B'(-2, 5)$. This transformation is represented by $\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} a \\ b \end{pmatrix}$. Find the values of a and b.

A a = 0, b = 5

B a = -5, b = 0 ✓

C a = 5, b = 0

D a = 0, b = -5

WORKED SOLUTION & VERIFICATION

Subtracting the original coordinates of $A(1, 2)$ from its image $A'(-4, 2)$ gives the translation vector:

$$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -4 - 1 \\ 2 - 2 \end{pmatrix} = \begin{pmatrix} -5 \\ 0 \end{pmatrix}.$$

Checking with $B(3, 5) \to B'(-2, 5)$ confirms $(-2 - 3, 5 - 5) = (-5, 0)$. Thus, $a = -5$ and $b = 0$.

11 MCQ

A survey was conducted on a group of 100 students regarding their preferences for two subjects, Mathematics (M) and Science (S). The results are summarized in the two-way table below, where some values are unknown. If a student is chosen at random from this group, the probability that the student prefers Mathematics but not Science, $P(M \cap \neg S)$, is equal to the probability that the student prefers Science but not Mathematics, $P(\neg M \cap S)$. What is the value of x?

	Science (S)	Not Science (¬S)	Total
Mathematics (M)	25	x	45
Not Math (¬M)	y	35	55
Total	45	z	100

A 30

B 15

C 20 ✓

D 25

WORKED SOLUTION & VERIFICATION

From the first row (Mathematics), the total number of students who prefer Mathematics is 45, so:

$$25 + x = 45 \implies x = 20.$$

Since $P(M \cap \neg S) = P(\neg M \cap S)$, we also have $y = x = 20$, which matches the first column total $25 + 20 = 45$ and the second row total $20 + 35 = 55$.

12 MCQ

Consider a geometric sequence with positive common ratio $r > 0$ where the first term is $a_1 = \frac{1}{2}$. If the sum of the first three terms is $S_3 = \frac{7}{2}$, find the value of r.

A -2

B 2 ✓

C 1/2

D 3

WORKED SOLUTION & VERIFICATION

The sum of the first three terms is $a_1(1 + r + r^2) = \frac{7}{2}$. Substituting $a_1 = \frac{1}{2}$ gives:

$$1 + r + r^2 = 7 \implies r^2 + r - 6 = (r + 3)(r - 2) = 0.$$

Since $r > 0$, the common ratio is $r = 2$.

13 MCQ

A geometric sequence has $a_1 = 2$ and common ratio $r = 3$. Find a_5.

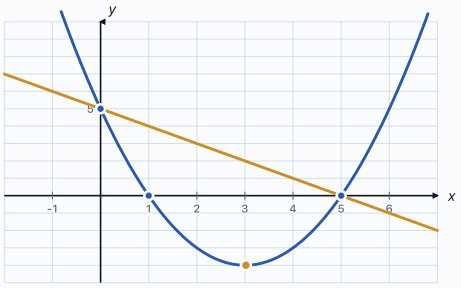
- **A** 162 ✓
- **B** 486
- **C** 165
- **D** -162

WORKED SOLUTION & VERIFICATION

Applying Sequences & Series principles, $a_5 = a_1 r^{n-1} = 2(3)^4 = 162$. Therefore, the verified correct option is 162.

14 MCQ

Consider the functions $f(x) = x^2 - 6x + 5$ and $g(x) = -x + 5$. The graphs of these two functions intersect at two points. Let these points be A and B. A third function, $h(x)$, is defined as the difference between the x-coordinates of these intersection points, i.e., $h(x) = |x_A - x_B|$. What is the value of $h(x)$?



- **A** $\sqrt{25}$ ✓
- **B** $\sqrt{16}$
- **C** $\sqrt{36}$
- **D** $\sqrt{49}$

WORKED SOLUTION & VERIFICATION

To find the intersection points of the graphs of $f(x)$ and $g(x)$, we set $f(x) = g(x)$.

$x^2 - 6x + 5 = -x + 5$

Subtract $-x + 5$ from both sides:

$x^2 - 6x + 5 - (-x + 5) = 0$

$x^2 - 6x + 5 + x - 5 = 0$

Combine like terms:

$x^2 - 5x = 0$

Factor out x:

$x(x - 5) = 0$

This gives two possible x-values for the intersection points:

$x_1 = 0$

$x_2 = 5$

Let $x_A = 0$ and $x_B = 5$ (or vice versa). The problem defines $h(x)$ as the absolute difference between these x-coordinates: $h(x) = |x_A - x_B|$.

$h(x) = |0 - 5| = |-5| = 5$.

We are asked for the value of $h(x)$, which is 5. We need to express this value in a form that matches the choices. The choices are given as square roots.

$\sqrt{16} = 4$

$\sqrt{25} = 5$

$\sqrt{36} = 6$

$\sqrt{49} = 7$

The value of $h(x)$ is 5, which corresponds to $\sqrt{25}$.

15 MCQ

In an applied problem, a geometric sequence has $a_1 = 6$ and common ratio $r = 3$. Find a_5.

A 1458

B 489

C -486

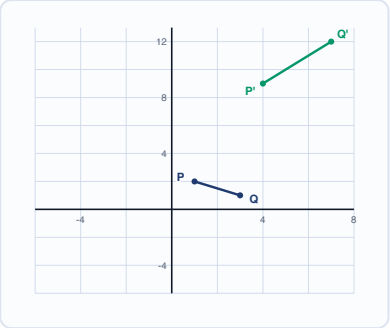
D **486** ✓

WORKED SOLUTION & VERIFICATION

Applying Sequences & Series principles, $a_5 = a_1 r^{n-1} = 6(3)^4 = 486$. Therefore, the verified correct option is 486.

16 MCQ

A linear transformation in the 2D plane is represented by the matrix $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. This transformation maps the point $P(1, 2)$ to $P'(-3, 7)$ and the point $Q(3, 1)$ to $Q'(5, 1)$. Determine the values of a, b, c, and d, and then calculate the sum $a + b + c + d$.



A 11

B 7

C 9 ✓

D 5

WORKED SOLUTION & VERIFICATION

Let the linear transformation matrix be $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

The transformation of a point $\mathbf{x}$ to $\mathbf{x}'$ is given by $M\mathbf{x} = \mathbf{x}'$.

For point $P(1, 2)$ mapped to $P'(4, 9)$:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \\ 9 \end{pmatrix}$$

This yields the equations:

1) $a + 2b = 4$

2) $c + 2d = 9$

For point $Q(3, 1)$ mapped to $Q'(7, 12)$:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 7 \\ 12 \end{pmatrix}$$

This yields the equations:

3) $3a + b = 7$

4) $3c + d = 12$

We need to solve these two systems of linear equations for a, b, c, and d.

Solving for a and b using equations (1) and (3):

From equation (3), $b = 7 - 3a$. Substitute this into equation (1):

$a + 2(7 - 3a) = 4$

$a + 14 - 6a = 4$

$-5a = 4 - 14$

$-5a = -10$

$a = 2$

Now substitute the value of a back into the expression for b:

$b = 7 - 3(2) = 7 - 6 = 1$

Solving for c and d using equations (2) and (4):

From equation (4), $d = 12 - 3c$. Substitute this into equation (2):

$c + 2(12 - 3c) = 9$

$c + 24 - 6c = 9$

$-5c = 9 - 24$

$-5c = -15$
$c = 3$
Now substitute the value of c back into the expression for d:
$d = 12 - 3(3) = 12 - 9 = 3$
So, the transformation matrix is $M = \begin{pmatrix} 2 & 1 \\ 3 & 3 \end{pmatrix}$.
The sum of the elements of the transformation matrix is $a + b + c + d = 2 + 1 + 3 + 3 = 9$.
Common misconceptions might arise from errors in solving the systems of equations, such as sign errors or algebraic mistakes when isolating variables. For instance, incorrectly solving for b or d could lead to a wrong sum. Alternatively, a student might forget to sum all four elements and only sum the elements of one row or column.

17 MCQ

A geometric sequence has $a_1 = 2$ and common ratio $r = -2$. Find a_4.

A 32	B **-16** ✓
C -18	D 16

WORKED SOLUTION & VERIFICATION
$a_4 = a_1 r^{n-1} = 2(-2)^3 = -16$.

18 MCQ

Suppose a geometric sequence has $a_1 = 5$ and common ratio $r = 3$. Find a_3.

A 135	B 48
C -45	D **45** ✓

WORKED SOLUTION & VERIFICATION
Applying Sequences & Series principles, $a_3 = a_1 r^{n-1} = 5(3)^2 = 45$. Therefore, the verified correct option is 45.

19 MCQ

Consider the complex number $z = 3 + 4i$. What is the magnitude of z?

A 7	B **5** ✓
C 12	D 25

WORKED SOLUTION & VERIFICATION
The magnitude of a complex number $z = a + bi$ is given by $|z| = \sqrt{a^2 + b^2}$. For $z = 3 + 4i$, we have $a = 3$ and $b = 4$. Therefore, the magnitude is $|z| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

20 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. What are the coordinates of the vertex of the parabola represented by this function?

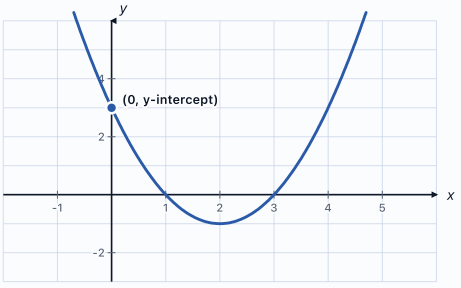
A (2, 1)	B (1, 0)
C (3, 0)	D **(2, -1)** ✓

WORKED SOLUTION & VERIFICATION

The given function is a quadratic function in the form $f(x) = ax^2 + bx + c$, where $a = 1$, $b = -4$, and $c = 3$. The x-coordinate of the vertex of a parabola is given by the formula $x = -\frac{b}{2a}$. Substituting the values, we get $x = -\frac{-4}{2(1)} = \frac{4}{2} = 2$. To find the y-coordinate of the vertex, we substitute this x-value back into the function: $f(2) = (2)^2 - 4(2) + 3 = 4 - 8 + 3 = -1$. Therefore, the coordinates of the vertex are (2, -1).

21 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. What is the y-intercept of the graph of this function?



A -3	B 4
C -4	D **3** ✓

WORKED SOLUTION & VERIFICATION

The y-intercept of a function's graph is the point where the graph crosses the y-axis. This occurs when $x = 0$. \nTo find the y-intercept, substitute $x = 0$ into the function: \n$f(0) = (0)^2 - 4(0) + 3$ \n$f(0) = 0 - 0 + 3$ \n$f(0) = 3$ \nTherefore, the y-intercept is 3.

22 MCQ

A geometric sequence has $a_1 = 1$ and common ratio $r = -3$. Find a_5.

A 78	B **81** ✓
C -81	D -243

WORKED SOLUTION & VERIFICATION

Applying Sequences & Series principles, $a_5 = a_1 r^{n-1} = 1(-3)^4 = 81$. Therefore, the verified correct option is 81.

23 MCQ

In an applied problem, a geometric sequence has $a_1 = 1$ and common ratio $r = -3$. Find a_6.

- **A** -243 ✓
- **B** -246
- **C** 243
- **D** 729

WORKED SOLUTION & VERIFICATION

$a_6 = a_1 r^{n-1} = 1(-3)^5 = -243$.

24 MCQ

An arithmetic sequence has its first term $a_1 = 7$ and a common difference $d = 5$. What is the value of the 3rd term, a_3?

- **A** 12
- **B** 17 ✓
- **C** 21
- **D** 35

WORKED SOLUTION & VERIFICATION

The formula for the nth term of an arithmetic sequence is $a_n = a_1 + (n-1)d$. \n Given $a_1 = 7$ and $d = 5$, we want to find a_3. \n Substitute $n = 3$ into the formula: \n $a_3 = 7 + (3-1) \times 5$ \n $a_3 = 7 + (2) \times 5$ \n $a_3 = 7 + 10$ \n $a_3 = 17$. \n Therefore, the 3rd term is 17.

25 MCQ

Let the complex number w be defined as $w = (a + bi) + (3 - 2i)$, where a and b are real numbers. Given that $|w|^2 = 25$ and w lies in the first quadrant, find the value of $a + b$.

- **A** 5
- **B** 7
- **C** 6 ✓
- **D** 8

WORKED SOLUTION & VERIFICATION

Let $w = (a + bi) + (3 - 2i) = (a + 3) + (b - 2)i$. We are given that $|w|^2 = 25$, so $(a + 3)^2 + (b - 2)^2 = 25$. Since w lies in the first quadrant, $a + 3 > 0$ and $b - 2 > 0$. We need to find integer or simple fractional solutions for $(a + 3)$ and $(b - 2)$ that sum of squares is 25. The possible pairs of squares are $(9, 16)$ and $(16, 9)$.
Case 1: $(a + 3)^2 = 9$ and $(b - 2)^2 = 16$. This gives $a + 3 = \pm 3$ and $b - 2 = \pm 4$. Since $a + 3 > 0$ and $b - 2 > 0$, we have $a + 3 = 3$ and $b - 2 = 4$. This leads to $a = 0$ and $b = 6$. However, if $a = 0$, then $w = 3 + 4i$, and $a + b = 0 + 6 = 6$.
Case 2: $(a + 3)^2 = 16$ and $(b - 2)^2 = 9$. This gives $a + 3 = \pm 4$ and $b - 2 = \pm 3$. Since $a + 3 > 0$ and $b - 2 > 0$, we have $a + 3 = 4$ and $b - 2 = 3$. This leads to $a = 1$ and $b = 5$. In this case, $w = 4 + 3i$, which is in the first quadrant. We want to find $a + b$. So, $a + b = 1 + 5 = 6$.
Both cases yield $a + b = 6$.